Bert vs. BioBert vs. Zero-Shot LLM in Medical NER

Tres passers (hopefully)
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1 Introduction and Context

This project compares a domain-specialist AI (BioBERT) against a generalist (Gemini) to see how they handle the shift from simple to complex medical language. We will categorize the BC5CDR dataset into "Easy" English (e.g., fever) and "Hard" Latinate (e.g., pyrexia) subsets. By measuring F1-scores, we can determine if specialized models fail due to word fragmentation during tokenization or if general AI hallucinates as terms become more obscure. This reveals exactly where big general knowledge outperforms targeted medical training.

2 Research Questions

- **RQ1:** Does BioBERT outperform Gemini on technical medical terms, and what type of errors does the respective models make?
- **RQ2:** How does sub-word fragmentation impact the models' ability to identify complex entities, and does performance degradation correlate strictly with word length or the number of tokenized fragments?

3 Datasets

- **BC5CDR**: PubMed abstracts annotated with diseases and chemicals. Used for fine-tuning BioBERT and evaluating all models.

4 Baseline Model

Our baseline is a standard BERT model (`bert-base-uncased`) trained on the EWT dataset. It is not exposed to any medical data, giving us a reference point before domain adaptation.

5 Methodology

We split the BC5CDR test set into two subsets based on word complexity:

- **Everyday terms** (e.g., *fever*, *rash*)
- **Technical terms** (e.g., *erythema*, *hepatotoxicity*)

We fine-tune BioBERT on BC5CDR. To directly address the generative nature of LLMs in an NER context, we deploy the following pipeline for Gemini via the AI Studio API:

- **Zero-Shot Prompting:** We instruct Gemini using a strict system prompt to act as an extraction engine without prior examples.
- **Structured Output & Parameters:** To avoid free-text generation, Gemini is prompted to return extractions exclusively in JSON format (e.g., `{"diseases": [...], "chemicals": [...]}`). We set the temperature to 0.0 to ensure deterministic outputs and minimize hallucinations.
- **Span Post-Processing:** Because LLMs struggle with character-level indexing, we map Gemini's JSON string outputs back to the original abstracts using exact string matching (via regex `re.finditer` in Python). This aligns the generative output with standard BIO-tagging formats, allowing a direct 1:1 F1-score comparison with BioBERT's span predictions.

6 Analysis Plan

- **Ablation:** BERT vs. BioBERT to isolate the effect of medical pre-training.
- **Token Evaluation:** For each model, we will evaluate how the number of sub-word fragments correlates with extraction performance. We will bucket entities into categories (e.g., 1 token, 2 tokens, 3+ tokens) and compare the F1-scores across these buckets while controlling for overall word length (character count) to isolate the effect of fragmentation.
- **Quantitative:** F1, Precision, and Recall per subset.
- **Qualitative:** Manual error inspection to compare failure types across models.

7 What Makes This Different

Most NER benchmarks report a single aggregate score. We argue this hides meaningful variation: a model can score well overall while completely failing on the rare, complex terms that matter most in clinical practice. By evaluating on different subsets—and conducting a deep dive into token-level fragmentation—we get a clearer view of where each approach is actually useful.